Analytical Cume

12 November 2010

The radiation for infrared spectroscopy is commonly obtained from a silicon carbide globar heated to 1,500 K by and electric current. The globar emits radiation with approximately the same spectrum as a blackbody radiator at 1,000 K.

$$M_\lambda = \frac{2\pi hc^2}{\lambda^5}\left(\frac{1}{e^{\frac{hc}{\lambda kT}} - 1}\right)$$

Where:

M_λ = radiant power per square meter per unit wavelength

h = Planck's constant = 6.626×10^{-34} J-s

c = speed of light 2.99×10^{8} m/s

λ = wavelength of radiation m

k = Boltzman's constant = 1.38×10^{-23} J/K

T = Temperature K

Where

1. Calculate M_λ at λ = 2.00 μm and at λ = 10.00 μm, at T = 1000 K.

   If you plot M_λ as a function of λ, you get a parabolic curve and the radiant power per unit area, P can be calculated by integrating the area under the curve between two narrowly selected wavelengths. If the wavelength range of $\Delta\lambda$ is small enough, M_λ can be considered constant between the two wavelengths and the power per unit area can be given by:

$$P = M_\lambda \Delta\lambda$$

2. Calculate the power emitted per square meter at 1000 K at 2.00 μm in the interval of λ = 1.99 μm to λ = 2.01 μm.
3. Repeat #2 for 10.00 μm for the interval λ = 9.99 μm to λ = 10.01 μm.
4. The quantity relative exitance, $E_{\text{-relative}}$, compares the power out at the two different wavelengths given by:

$$E_relative = \frac{M_\lambda(\lambda = 2\mu m)}{M_\lambda(\lambda = 10\mu m)}$$

Compare the relative exitance at these two wavelengths at 1000K with the relative exitance at 100 K. What does your answer mean?